

Trainings ▾

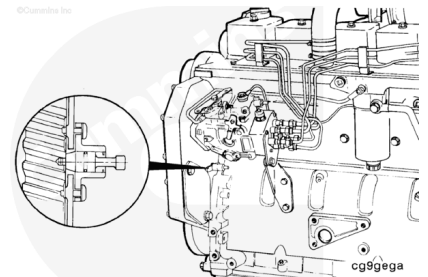
General Information

The timing pin is used to accurately locating TDC for:

- Fuel pump timing
- Overhead Set

The timing pin is typically located below the fuel pump:

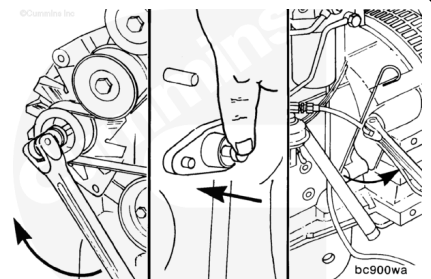
- For front gear train engines, in the front gear housing (shown)
- For rear gear train engines, in the rear gear housing (**not** shown).



To use the timing pin to locate top dead center (TDC) for cylinder number 1, bar the crankshaft slowly while pressing on the engine timing pin. Use the barring tool, Part Number 3824591, to bar the engine.

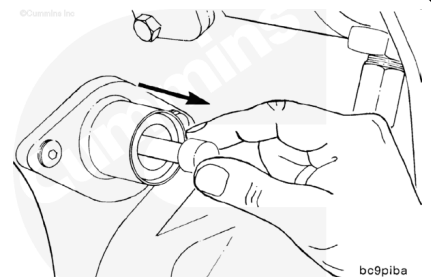
Bar the engine over until the timing pin engages the notch in the camshaft gear.

Service Tip: If the barring tool is inaccessible, in some applications the alternator pulley can be used to bar the engine over.



⚠ CAUTION ⚠

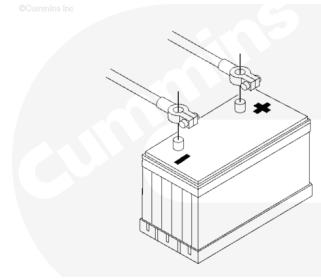
To reduce the possibility of engine or timing pin damage, you must disengage the timing pin before attempting to bar or crank the engine.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



13900050

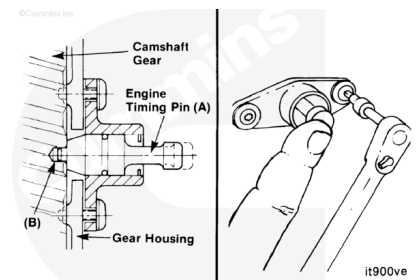
Disconnect the batteries.

Remove

Note : If the gear housing/gear train has not been removed, prior to removing the timing pin assembly, rotate the engine so that the timing pin engages the camshaft gear. See the General Information section of this procedure.

Remove the timing pin housing assembly.

Note : Most timing pin assemblies are mounted using internal Torx fasteners.



it900ve

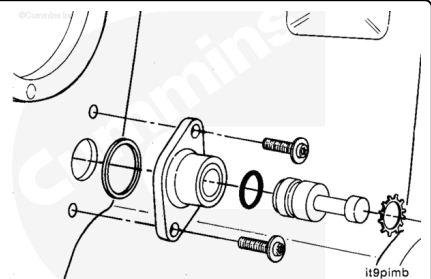
Install

The following steps **must** be performed before installing the timing pin assembly if one of the following has occurred:

- The gear housing was replaced
- The timing pin assembly was replaced with the gear housing removed
- The timing pin is suspected to be inaccurate.

The timing pins assembly is precisely located on the gear housing to correspond to TDC for cylinder number 1 on the compression stroke.

If only the timing pin assembly is being replaced, with the gear housing still installed, skip the following TDC locating steps.



it9pimb

Note : The following steps have been generalized to cover both front and rear gear train engines. Some of

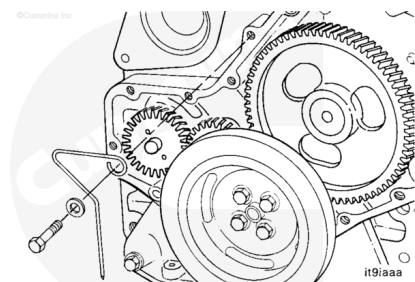
the illustrations may differ from the engine being serviced.

TDC **must** accurately be located using one of the following two methods:

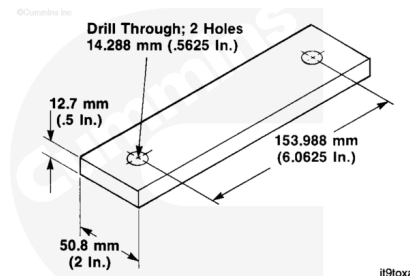
- If the cylinder head is removed, use a steel plate
- If the cylinder head is installed, use the engine's intake valve.

First, fabricate a pointer as illustrated. Attached the pointer to the front gear cover/gear housing using flat washers to prevent damage to the front gear cover/housing.

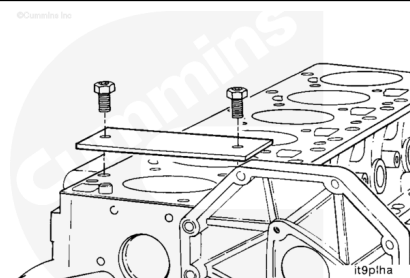
Note : If the vibration damper/crankshaft pulley has been removed, the vibration damper/crankshaft pulley **must** be temporarily installed.



If the cylinder head is **not** installed, fabricate a steel plate as shown in the illustration.

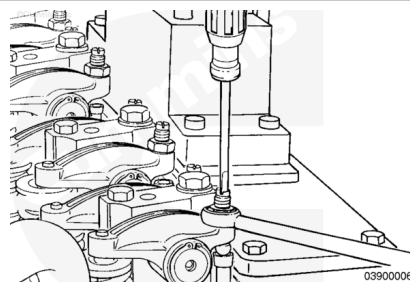


Using two correct thread and pitch capscrews, attach the plate over cylinder number 1.



⚠ CAUTION ⚠

Once the intake valve(s) are protruding into the cylinder bore, bar the engine over with ease. Too hard of contact with the intake valve(s) will bend the valves and cause serious engine damage.



If the cylinder head is installed, remove the rocker lever cover to gain access to the number 1 cylinder intake and exhaust rocker levers. Refer to Procedure 003-011 (</qs3/pubsys2/xml/en/procedures/40/40-003-011-tr.html>).

Use the barring tool, part number 3824591, rotate the engine over until cylinder number 1 exhaust valve(s) are fully open. The exhaust valve(s) are fully open when the exhaust valve spring(s) are compressed.

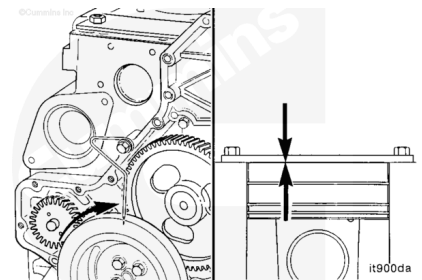
With the exhaust valve(s) fully open, set the overhead on the cylinder number 1 intake valve(s) so that the intake valve spring(s) are compressed.

Note : It may be necessary to bottom out the adjusting screw to achieve enough intake valve protrusion to contact the piston.

Note : The following steps will show illustrations of engines with the cylinder head removed. The following steps are the same for an engine with or without the cylinder head removed.

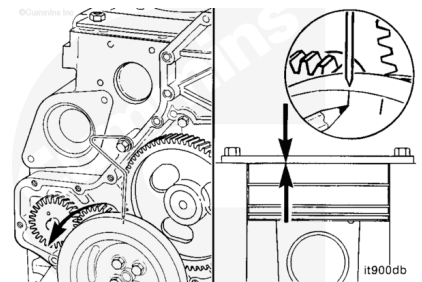
Use the barring tool, Part Number 3824591, rotate the engine clockwise until the piston makes contact with the steel plate or intake valve(s).

Mark the vibration damper/crankshaft pulley where the fabricated indicator points.

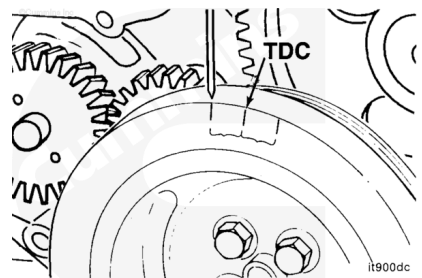


Use the barring tool, Part Number 3824591, rotate the engine counterclockwise until the piston makes contact with the plate or intake valve(s).

Mark the vibration damper/crankshaft pulley where the fabricated indicator points.



Place a mark on the vibration damper/crankshaft pulley at one-half the distance between the two marks. This is TDC for cylinder number one.

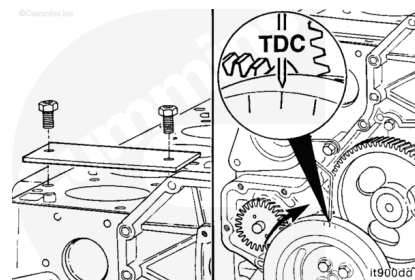


For engines with the cylinder head removed, remove the plate over cylinder number 1.

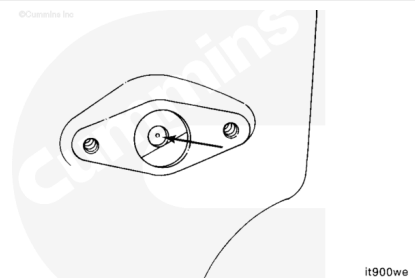
For engines with the cylinder head installed, adjust the intake valve(s) in order to close the valve(s).

Note : When rotating the engine on engines with the cylinder head installed, carefully watch the rocker levers and push rods. The overhead is **not** set for the number one intake valve(s) which may cause the push rod to lose contact with the rocker lever.

Use the barring tool, part number 3824591, rotate the engine until the pointer aligns with the TDC mark on the vibration damper/crankshaft pulley.



Look through the timing pin hole in the gear housing. Look for the timing pin hole in the camshaft gear. If it is **not** visible, rotate the crankshaft 360 degrees and align the pointer with the TDC mark.

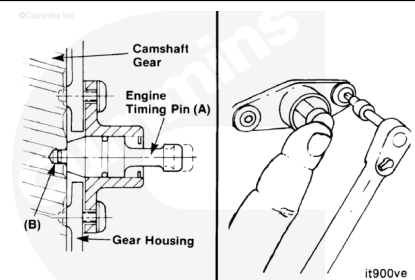


Install the timing pin assembly. Push the timing pin into the hole in the camshaft gear to align the housing.

While holding the timing pin, tighten the mounting capscrews.

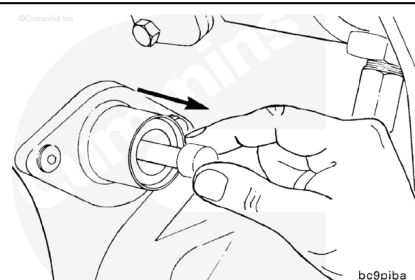
Torque Value: 5 n·m [48 in-lb]

Note : Most timing pin assemblies are mounted using internal Torx fasteners.



⚠ CAUTION ⚠

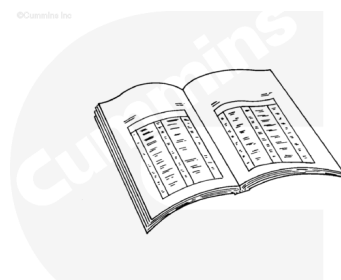
To reduce the possibility of engine or timing pin damage, you must disengage the timing pin before attempting to bar or crank the engine.



Finishing Steps

- For engines with the cylinder head installed, adjust the overhead for the intake valve(s) on cylinder number 1. Refer to Procedure 003-004 (</qs3/pubsys2/xml/en/procedures/40/40-003-004-tr.html>).

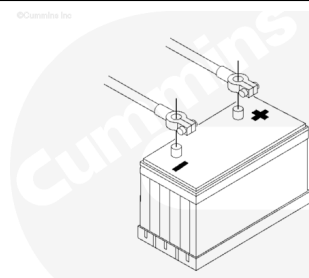
- For engines with the cylinder head installed, install the rocker lever cover. Refer to Procedure 003-011 (</qs3/pubsys2/xml/en/procedures/40/40-003-011-tr.html>).



ck800wa

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



13900050

Connect the batteries.